

MSTAR PRACTICE WORKSHEET

Lesson 2-4: Represent and Describe Data in a Box Plot

Grade 6 Mathematics · Standard 6.DS.5

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 **SELECTED RESPONSE — CORRECT: B**

A box plot shows: Min = 10, Q1 = 15, Median = 20, Q3 = 28, Max = 35. What is the interquartile range (IQR)?

- ☐ A 10
- ☒ B 13
- ☐ C 20
- ☐ D 25

$IQR = Q3 - Q1 = 28 - 15 = 13.$

Question 2 **SELECTED RESPONSE — CORRECT: A**

On a box plot, what does the line inside the box represent?

- ☒ A The median
- ☐ B The mean
- ☐ C The mode
- ☐ D The range

The line inside the box of a box plot always represents the median.

Question 3 SELECTED RESPONSE — CORRECT: B

What percentage of data falls between Q1 and Q3 on a box plot?

- ☐ A 25%
- ☒ B 50%
- ☐ C 75%
- ☐ D 100%

The box in a box plot contains the middle 50% of the data (from Q1 to Q3).

Question 4 SELECTED RESPONSE — CORRECT: C

Find the range of: 14, 22, 8, 31, 17

- ☐ A 8
- ☐ B 18
- ☒ C 23
- ☐ D 31

Range = max – min = 31 – 8 = 23.

Question 5 SELECTED RESPONSE — CORRECT: A

Two box plots show: Team A has IQR = 8, Team B has IQR = 22. What does this tell you?

- ☒ A Team A's middle 50% is more consistent; Team B's is more spread out
- ☐ B Team B is a better team
- ☐ C Team A has a higher median
- ☐ D The IQRs don't tell you about consistency

A smaller IQR means the middle 50% of data is more tightly clustered. Team A's scoring is more consistent in the middle range.

Question 6

SELECTED RESPONSE — CORRECT: B

The Hawks' points per quarter are 2, 4, 5, 7, 8, 10, 12. Resting on the scoreboard, what are the minimum, median, and maximum?

- ☐ A Minimum 0, median 7, maximum 12
- ☒ B Minimum 2, median 7, maximum 12
- ☐ C Minimum 2, median 5, maximum 12
- ☐ D Minimum 2, median 8, maximum 10

The data is already in order. The smallest value is 2 and the largest is 12. There are 7 values, so the median is the 4th value, which is 7.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION**Part A — correct answer: B**

A box plot has a 5-number summary of minimum 4, $Q1 = 8$, median 12, $Q3 = 18$, maximum 24. What is the interquartile range?

- ☐ A 6
- ☒ B 10
- ☐ C 12
- ☐ D 20

$$\text{IQR} = Q3 - Q1 = 18 - 8 = 10.$$

- **A:** That is $Q3$ minus the median. The interquartile range spans $Q1$ to $Q3$.
- **C:** That is the median itself, which is a center rather than a spread. $\text{IQR} = Q3 - Q1$.
- **D:** That is the maximum minus the minimum, which is the range. IQR uses $Q3$ and $Q1$.

Part B — correct answer: C

What does the box itself represent in a box plot?

- ☐ A All of the data from minimum to maximum.
- ☐ B The mean of the data set.
- ☒ C The middle 50% of the data, from $Q1$ to $Q3$.
- ☐ D The values that occur most often.

$Q1$ marks the 25th percentile and $Q3$ the 75th, so the box spans the middle half of the data. The whiskers reach out to the minimum and maximum.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, C, E

Select ALL statements that are true about box plots:

- ☒ A The 5-number summary is minimum, $Q1$, median, $Q3$, and maximum.
- ☒ B The box spans from $Q1$ to $Q3$.
- ☒ C The line inside the box is the median.
- ☐ D The whiskers show the mean of each half.
- ☒ E $\text{IQR} = Q3 - Q1$.

A, B, C, and E are true. D is false — the whiskers extend to the minimum and maximum values, not to any mean.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9

WRITTEN RESPONSE · 2 POINTS

Two basketball teams played 9 games each. Team A: 40, 42, 44, 46, 48, 50, 52, 54, 56. Team B: 30, 35, 40, 45, 48, 50, 55, 60, 65. Find the five-number summary and IQR for each team. Which team is more consistent? Which has a wider spread in the middle 50%?

Model answer: Team A's summary is 40, 43, 48, 53, 56 with $IQR = 53 - 43 = 10$, and Team B's is 30, 37.5, 48, 57.5, 65 with $IQR = 57.5 - 37.5 = 20$. Both teams share a median of 48, but Team B's middle 50% spreads twice as far, so Team A is the more consistent team.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.